

Bài toán 1

Giả sử f' liên tục tuyệt đối và $\int (f'(u))^2 du < \infty$. Khi đó

$$R(\hat{f}_n, f) = \frac{h^2}{12} \int (f'(u))^2 du + \frac{1}{nh} + o(h^2) + o\left(\frac{1}{n}\right).$$

$R(\hat{f}_n, f)$ đạt giá trị nhỏ nhất tại h^* với

$$h^* = \frac{1}{n^{1/3}} \left(\frac{6}{\int (f'(u))^2 du} \right)^{1/3}.$$

Ngoài ra,

$$R(\hat{f}_n, f) \sim \frac{C}{n^{2/3}}$$

trong đó $C = (3/4)^{2/3} \left(\int (f'(u))^2 du \right)^{1/3}$.

Lời giải. Ta có

$$R(\hat{f}_n, f) = \int b^2(x) dx + \int \text{Var}[\hat{f}_n(x)].$$

* Tính $\int b^2(x) dx$:

Với bất kì $x, u \in B_j$, ta có

$$f(u) = f(x) + (u - x)f'(x) + \frac{(u - x)^2}{2} f''(\tilde{x})$$

với $x < \tilde{x} < u$. Do đó,

$$\begin{aligned} p_j &= \int_{B_j} f(u) du = \int_{B_j} \left(f(x) + (u - x)f'(x) + \frac{(u - x)^2}{2} f''(\tilde{x}) \right) du \\ &= f(x)h + hf'(x) \left(h \left(j - \frac{1}{2} \right) - x \right) + O(h^3). \end{aligned}$$

Cho nên

$$\begin{aligned} b(x) &= \mathbb{E}[\hat{f}_n(x)] - f(x) = \frac{p_j}{h} - f(x) \\ &= \frac{f(x)h + hf'(x) \left(h \left(j - \frac{1}{2} \right) - x \right) + O(h^3)}{h} - f(x) \\ &= f'(x) \left(h \left(j - \frac{1}{2} \right) - x \right) + O(h^2) \end{aligned}$$

Áp dụng định lý giá trị trung bình, với $\tilde{x}_j \in B_j$, ta được

$$\begin{aligned}\int_{B_j} b^2(x) dx &= \int_{B_j} (f'(x))^2 \left(h \left(j - \frac{1}{2} \right) - x \right)^2 dx + O(h^4) \\ &= (f'(\tilde{x}_j))^2 \int_{B_j} \left(h \left(j - \frac{1}{2} \right) - x \right)^2 dx + O(h^4) \\ &= (f'(\tilde{x}_j))^2 \frac{h^3}{12} + O(h^4)\end{aligned}$$

Cho nên

$$\begin{aligned}\int b^2(x) dx &= \int_0^1 b^2(x) dx = \sum_{j=1}^m \int_{B_j} b^2(x) dx + O(h^3) \\ &= \sum_{j=1}^m (f'(\tilde{x}_j))^2 \frac{h^3}{12} + O(h^3) \\ &= \frac{h^2}{12} \sum_{j=1}^m h (f'(\tilde{x}_j))^2 + O(h^3) \\ &= \frac{h^2}{12} \int_0^1 (f'(x))^2 dx + o(h^2).\end{aligned}$$

* Tính $\int \text{Var} [\hat{f}_n(x)]:$

Cũng áp dụng định lý giá trị trung bình, với $x_j \in B_j$, ta được $p_j = \int_{B_j} f(x) dx = hf(x_j)$. Do đó

$$\begin{aligned}\int \text{Var} [\hat{f}_n(x)] &= \int_0^1 \text{Var} [\hat{f}_n(x)] dx = \sum_{j=1}^m \int_{B_j} \text{Var} [\hat{f}_n(x)] dx = \sum_{j=1}^m \int_{B_j} \frac{p_j(1-p_j)}{nh^2} \\ &= \frac{1}{nh^2} \sum_{j=1}^m \int_{B_j} p_j - \frac{1}{nh^2} \sum_{j=1}^m \int_{B_j} p_j^2 = \frac{1}{nh^2} \sum_{j=1}^m hp_j - \frac{1}{nh^2} \sum_{j=1}^m \int_{B_j} hp_j^2 \\ &= \frac{1}{nh} \sum_{j=1}^m p_j - \frac{1}{nh} \sum_{j=1}^m p_j^2 = \frac{1}{nh} - \frac{1}{nh} \sum_{j=1}^m p_j^2 \\ &= \frac{1}{nh} - \frac{1}{nh} \sum_{j=1}^m h^2 f^2(x_j) = \frac{1}{nh} - \frac{1}{n} \sum_{j=1}^m hf^2(x_j) \\ &= \frac{1}{nh} - \frac{1}{n} \left(\int_0^1 f^2(x) dx + o(1) \right) = \frac{1}{nh} + o\left(\frac{1}{n}\right).\end{aligned}$$

Vậy ta được

$$R(\hat{f}_n, f) = \int b^2(x) dx + \int \text{Var} [\hat{f}_n(x)] = \frac{h^2}{12} \int (f'(u))^2 du + \frac{1}{nh} + o(h^2) + o\left(\frac{1}{n}\right).$$

Ta có

$$\frac{\partial R(\hat{f}_n, f)}{\partial h} = \frac{h}{6} \int (f'(x))^2 dx - \frac{1}{nh^2} + o(h).$$

Suy ra

$$\frac{\partial R(\hat{f}_n, f)}{\partial h} = 0 \Leftrightarrow \frac{h}{6} \int (f'(x))^2 dx - \frac{1}{nh^2} + o(h) = 0,$$

hay

$$1 = nh^3 o(1) + \frac{nh^3}{6} \int (f'(x))^2 dx = nh^3 \left(\frac{1}{6} \int (f'(x))^2 dx + o(1) \right) = nh^3 \left(\frac{1}{6} \int (f'(x))^2 dx \right).$$

Do đó, ta được

$$h^* = \frac{1}{n^{1/3}} \left(\frac{6}{\int (f'(u))^2 du} \right)^{1/3}.$$

Bài toán 2

Chứng minh

$$\hat{J}(h) = \frac{2}{h(n-1)} - \frac{n+1}{h(n-1)} \sum_{j=1}^m \hat{p}_j^2.$$

Lời giải. Ta có

$$\hat{p}_j = \frac{1}{n} \sum_{i=1}^n I(X_i \in B_j).$$

Suy ra

$$\hat{f}_n(x) = \sum_{j=1}^m \frac{\hat{p}_j}{h} I(x \in B_j) = \frac{1}{nh} \sum_{j=1}^m \sum_{k=1}^n I(X_k \in B_j) I(x \in B_j).$$

Do đó

$$\begin{aligned} \hat{f}_{(-i)}(X_i) &= \frac{1}{(n-1)h} \sum_{j=1}^m \sum_{k=1, k \neq i}^n I(X_k \in B_j) I(X_i \in B_j) \\ &= \frac{1}{(n-1)h} \sum_{j=1}^m (n\hat{p}_j - I(X_i \in B_j)) I(X_i \in B_j) \\ &= \frac{1}{(n-1)h} \left[\sum_{j=1}^m n\hat{p}_j I(X_i \in B_j) - \sum_{j=1}^m I(X_i \in B_j) \right] \end{aligned}$$

Suy ra

$$\begin{aligned} \frac{2}{n} \sum_{i=1}^n \hat{f}_{(-i)}(X_i) &= \frac{2}{n(n-1)h} \sum_{i=1}^n \left[\sum_{j=1}^m n\hat{p}_j I(X_i \in B_j) - \sum_{j=1}^m I(X_i \in B_j) \right] \\ &= \frac{2n}{(n-1)h} \sum_{j=1}^m \hat{p}_j \left(\frac{1}{n} \sum_{i=1}^n I(X_i \in B_j) \right) - \frac{2}{(n-1)h} \sum_{j=1}^m \left(\frac{1}{n} \sum_{i=1}^n I(X_i \in B_j) \right) \\ &= \frac{2n}{(n-1)h} \sum_{j=1}^m \hat{p}_j^2 - \frac{2}{(n-1)h} \sum_{j=1}^m \hat{p}_j = \frac{2n}{(n-1)h} \sum_{j=1}^m \hat{p}_j^2 - \frac{2}{(n-1)h} \end{aligned}$$

Tiếp theo, ta có

$$\begin{aligned}
\int \left(\hat{f}_n(x) \right)^2 dx &= \frac{1}{n^2 h^2} \int_0^1 \left[\sum_{j=1}^m \sum_{k=1}^n I(X_k \in B_j) I(x \in B_j) \right]^2 dx \\
&= \frac{1}{n^2 h^2} \sum_{l=1}^m \int_{B_l} \left[\sum_{j=1}^m \sum_{k=1}^n I(X_k \in B_j) I(x \in B_j) \right]^2 dx \\
&= \frac{1}{n^2 h^2} \sum_{l=1}^m \int_{B_l} \left[\sum_{k=1}^n I(X_k \in B_l) \right]^2 dx = \frac{1}{n^2 h^2} \sum_{l=1}^m \int_{B_l} (n \hat{p}_l)^2 dx \\
&= \frac{1}{h^2} \sum_{l=1}^m \hat{p}_l^2 \int_{B_l} dx = \frac{1}{h} \sum_{l=1}^m \hat{p}_l^2 = \frac{1}{h} \sum_{j=1}^m \hat{p}_j^2.
\end{aligned}$$

Từ đó, ta được

$$\begin{aligned}
\hat{J}(h) &= \int \left(\hat{f}_n(x) \right)^2 dx - \frac{2}{n} \sum_{i=1}^n \hat{f}_{(-i)}(X_i) \\
&= \frac{1}{h} \sum_{j=1}^m \hat{p}_j^2 - \frac{2n}{(n-1)h} \sum_{j=1}^m \hat{p}_j^2 + \frac{2}{(n-1)h} \\
&= \frac{2}{h(n-1)} - \frac{n+1}{h(n-1)} \sum_{j=1}^m \hat{p}_j^2.
\end{aligned}$$